ARTIFICIAL INTELLIGENCE GOVERNANCE AND INDIAAI

AI TAXONOMY LADDER → CAPABILITY-TO-DEPLOYMENT LADDER → FOUNDATION-MODEL RISK TREE → INDIAAI INSTITUTION MAP → MISSION PILLAR WHEEL → COMPUTE MEASUREMENT FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ● SUBORDINATE EXTRA ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

00

STAGE 00 AI TAXONOMY LADDER

AI TAXONOMY LADDER · ARTIFICIAL INTELLIGENCE · BROAD TASK-PERFORMING SYSTEMS · MACHINE LEARNING · PATTERNS LEARNED FROM DATA · DEEP LEARNING · MULTILAYER NEURAL NETWORKS · GENERATIVE AI

ARTIFICIAL INTELLIGENCE → broad task-performing systems

MACHINE LEARNING → patterns learned from data → **DEEP LEARNING → multilayer neural networks** → **GENERATIVE AI → creates text, image, audio or code**

STARTING CONDITION / INPUT
- ARTIFICIAL INTELLIGENCE → broad task-performing systems
- MACHINE LEARNING → patterns learned from data

SCIENTIFIC OR ENGINEERING MECHANISM
- DEEP LEARNING → multilayer neural networks
- GENERATIVE AI → creates text, image, audio or code

OUTPUT / FAILURE BOUNDARY
- RULE → identify method, task and evidence before impact
- ARTIFICIAL INTELLIGENCE → broad task-performing systems

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → identify method, task and evidence before impact.

01

STAGE 01 CAPABILITY-TO-DEPLOYMENT LADDER

CAPABILITY-TO-DEPLOYMENT LADDER · RESEARCH CLAIM · PROPOSED METHOD · BENCHMARK RESULT · BOUNDED EVALUATION · INTEGRATED DEMONSTRATION · REAL WORKFLOW WITH USERS AND MONITORING · MEASURED BENEFIT AND HARM AFTER USE

RESEARCH CLAIM → proposed method

BENCHMARK RESULT → bounded evaluation → **PROTOTYPE → integrated demonstration** → **DEPLOYMENT → real workflow with users and monitoring**

STARTING CONDITION / INPUT
- RESEARCH CLAIM → proposed method
- BENCHMARK RESULT → bounded evaluation

SCIENTIFIC OR ENGINEERING MECHANISM
- PROTOTYPE → integrated demonstration
- DEPLOYMENT → real workflow with users and monitoring

OUTPUT / FAILURE BOUNDARY
- OUTCOME → measured benefit and harm after use
- RESEARCH CLAIM → proposed method

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → measured benefit and harm after use.

02

STAGE 02 FOUNDATION-MODEL RISK TREE

FOUNDATION-MODEL RISK TREE · PRETRAINED FOUNDATION MODEL · BROAD DOWNSTREAM REUSE · PROVENANCE CONSTRAINTS · INHERITED APPLICATION RISK · FLUENT OUTPUT WITHOUT VERIFICATION · GROUNDING, EVALUATION AND HUMAN REVIEW

PRETRAINED FOUNDATION MODEL → broad downstream reuse

DATA / LICENCE → provenance constraints → **BIAS / SAFETY → inherited application risk** → **HALLUCINATION → fluent output without verification**

CONCEPT / ARCHITECTURE
- PRETRAINED FOUNDATION MODEL → broad downstream reuse
- DATA / LICENCE → provenance constraints

MECHANISM / APPLICATION
- BIAS / SAFETY → inherited application risk
- HALLUCINATION → fluent output without verification

READINESS / STATUS LIMIT
- CONTROL → grounding, evaluation and human review
- PRETRAINED FOUNDATION MODEL → broad downstream reuse

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HALLUCINATION → fluent output without verification; CONTROL → grounding, evaluation and human review.

03

STAGE 03 INDIAAI INSTITUTION MAP

INDIAAI INSTITUTION MAP · MINISTRY POLICY AND RULE-MAKING ANCHOR · DIGITAL INDIA CORPORATION · MISSION IMPLEMENTATION · NITI AAYOG · EARLIER STRATEGY AND THINK-TANK ROLE · SECTOR REGULATOR · DEPLOYMENT OBLIGATIONS

MeitY → ministry policy and rule-making anchor

INDIAAI / DIGITAL INDIA CORPORATION → mission implementation → **NITI AAYOG → earlier strategy and think-tank role** → **SECTOR REGULATOR / DEPARTMENT → deployment obligations**

CONCEPT / ARCHITECTURE
- MeitY → ministry policy and rule-making anchor
- INDIAAI / DIGITAL INDIA CORPORATION → mission implementation

MECHANISM / APPLICATION
- NITI AAYOG → earlier strategy and think-tank role
- SECTOR REGULATOR / DEPARTMENT → deployment obligations

READINESS / STATUS LIMIT
- RULE → strategy, mission, regulator and law stay separate
- MeitY → ministry policy and rule-making anchor

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → strategy, mission, regulator and law stay separate.

04

STAGE 04 MISSION PILLAR WHEEL

MISSION PILLAR WHEEL · COMPUTE + INNOVATION CENTRE + APPLICATIONS · AIKOSH + STARTUP FINANCING + FUTURESKILLS · SAFE AND TRUSTED AI · GOVERNANCE TOOLS AND EVALUATION · IMPLEMENTATION DOMAIN · PILLAR NAME IS NOT COMPLETED OUTPUT

COMPUTE + INNOVATION CENTRE + APPLICATIONS

AIKOSH + STARTUP FINANCING + FUTURESKILLS → **SAFE AND TRUSTED AI → governance tools and evaluation** → **PILLAR → implementation domain**

CONCEPT / ARCHITECTURE
- COMPUTE + INNOVATION CENTRE + APPLICATIONS
- AIKOSH + STARTUP FINANCING + FUTURESKILLS

MECHANISM / APPLICATION
- SAFE AND TRUSTED AI → governance tools and evaluation
- PILLAR → implementation domain

READINESS / STATUS LIMIT
- TRAP → pillar name is not completed output
- COMPUTE + INNOVATION CENTRE + APPLICATIONS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → pillar name is not completed output.

05

STAGE 05 COMPUTE MEASUREMENT FIREWALL

COMPUTE MEASUREMENT FIREWALL · GPU OBJECTIVE · PLANNED HARDWARE MEASURE · EMPANELLED CAPACITY · PROCUREMENT AVAILABILITY · COMPUTE UNIT · PROVIDER-DEFINED SERVICE MEASURE · USED CAPACITY

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
GPU OBJECTIVE → planned hardware measure	COMPUTE UNIT → provider-defined service measure	RULE → never convert one measure into another
EMPANELLED CAPACITY → procurement availability	BOOKED / USED CAPACITY → actual access or consumption	GPU OBJECTIVE → planned hardware measure

SYSTEM / DEVICE / INSTITUTION
- GPU OBJECTIVE → planned hardware measure
- EMPANELLED CAPACITY → procurement availability

DECISIVE TECHNICAL DISTINCTION
- COMPUTE UNIT → provider-defined service measure
- BOOKED / USED CAPACITY → actual access or consumption

STATUS / UNIT / EVIDENCE LIMIT
- RULE → never convert one measure into another
- GPU OBJECTIVE → planned hardware measure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → never convert one measure into another.

06

STAGE 06 AIKOSH EVIDENCE GATE

AIKOSH EVIDENCE GATE · DATASET LISTING · LAWFUL ORIGIN AND PERMITTED USE · FITNESS FOR INDIAN CONTEXT · CONTROLLED USE · PLATFORM PRESENCE ALONE PROVES NONE OF THESE

DATASET LISTING → discoverability → **PROVENANCE / LICENCE → lawful origin and permitted use** → **QUALITY / REPRESENTATION → fitness for Indian context** → **ACCESS / SECURITY → controlled use** → **OUTCOME → platform presence alone proves none of these**

STARTING CONDITION / INPUT
- DATASET LISTING → discoverability
- PROVENANCE / LICENCE → lawful origin and permitted use

SCIENTIFIC OR ENGINEERING MECHANISM
- QUALITY / REPRESENTATION → fitness for Indian context
- ACCESS / SECURITY → controlled use

OUTPUT / FAILURE BOUNDARY
- OUTCOME → platform presence alone proves none of these
- DATASET LISTING → discoverability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → platform presence alone proves none of these.

07

STAGE 07 USE-CASE DEPLOYMENT CHAIN

USE-CASE DEPLOYMENT CHAIN · PUBLIC PROBLEM · DEFINE LEGITIMATE OBJECTIVE · DATA + MODEL · BUILD OR PROCURE COMPONENT · INTERFACE + HUMAN WORKFLOW · OPERATIONAL DECISION · MONITORING + APPEAL

PUBLIC PROBLEM → define legitimate objective → **DATA + MODEL → build or procure component** → **INTERFACE + HUMAN WORKFLOW → operational decision** → **MONITORING + APPEAL → accountability after output** → **RULE → model capability is not service outcome**

STARTING CONDITION / INPUT
- PUBLIC PROBLEM → define legitimate objective
- DATA + MODEL → build or procure component

SCIENTIFIC OR ENGINEERING MECHANISM
- INTERFACE + HUMAN WORKFLOW → operational decision
- MONITORING + APPEAL → accountability after output

OUTPUT / FAILURE BOUNDARY
- RULE → model capability is not service outcome
- PUBLIC PROBLEM → define legitimate objective

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → model capability is not service outcome.

08

STAGE 08 RISK LIFECYCLE

RISK LIFECYCLE · PROBLEM SELECTION · EXCLUSION OR MISSION CREEP · BIAS, PRIVACY AND PROVENANCE · HIDDEN FAILURE OR LOCK-IN · AUTOMATION BIAS AND DRIFT · IMPACT ASSESSMENT, LOGS, INCIDENTS AND REVIEW

PROBLEM SELECTION → exclusion or mission creep → **DATA / TRAINING → bias, privacy and provenance** → **EVALUATION / PROCUREMENT → hidden failure or lock-in** → **DEPLOYMENT / UPDATE → automation bias and drift** → **CONTROL → impact assessment, logs, incidents and review**

STARTING CONDITION / INPUT
- PROBLEM SELECTION → exclusion or mission creep
- DATA / TRAINING → bias, privacy and provenance

SCIENTIFIC OR ENGINEERING MECHANISM
- EVALUATION / PROCUREMENT → hidden failure or lock-in
- DEPLOYMENT / UPDATE → automation bias and drift

OUTPUT / FAILURE BOUNDARY
- CONTROL → impact assessment, logs, incidents and review
- PROBLEM SELECTION → exclusion or mission creep

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONTROL → impact assessment, logs, incidents and review.

09

STAGE 09 RISK AND CONTROL MATRIX

RISK AND CONTROL MATRIX · REPRESENTATIVE TESTS AND REMEDY · LAWFUL PURPOSE AND MINIMISATION · ACCESS CONTROL AND RESILIENCE · EVALUATION, EXPLANATION AND ESCALATION · PROVENANCE, LABELLING AND ENFORCEMENT · MUST REMEMBER · AI ANALYSIS SEPARATES MODEL, TRAINING/INFERENCE, DATA, APPLICATION

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
BIAS → representative tests and remedy	SECURITY → access control and resilience	MISINFORMATION → provenance, labelling and enforcement
PRIVACY → lawful purpose and minimisation	SAFETY / OPACITY → evaluation, explanation and escalation	MUST REMEMBER: AI analysis separates model, training/inference, data, application,...

SYSTEM / DEVICE / INSTITUTION
- BIAS → representative tests and remedy
- PRIVACY → lawful purpose and minimisation

DECISIVE TECHNICAL DISTINCTION
- SECURITY → access control and resilience
- SAFETY / OPACITY → evaluation, explanation and escalation

STATUS / UNIT / EVIDENCE LIMIT
- MISINFORMATION → provenance, labelling and enforcement
- MUST REMEMBER: AI analysis separates model, training/inference, data, application,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: AI analysis separates model, training/inference, data, application,.

10

STAGE 10 INSTRUMENT HIERARCHY

INSTRUMENT HIERARCHY · PERSUASIVE COMMUNICATION · CONSULTATION REPORT · NON-BINDING FRAMEWORK · NOTIFIED RULE · BINDING UNDER PARENT ACT · PRIMARY LEGISLATION · SECTORAL LAW

ADVISORY → persuasive communication

CONSULTATION REPORT / GUIDELINE → non-binding framework → **NOTIFIED RULE → binding under parent Act** → **ACT → primary legislation**

STARTING CONDITION / INPUT
- ADVISORY → persuasive communication
- CONSULTATION REPORT / GUIDELINE → non-binding framework

SCIENTIFIC OR ENGINEERING MECHANISM
- NOTIFIED RULE → binding under parent Act
- ACT → primary legislation

OUTPUT / FAILURE BOUNDARY
- SECTORAL LAW → context-specific enforceable duty
- CLOSE DISTINCTION: Model is not application, benchmark is not real-world safety, compute...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Model is not application, benchmark is not real-world safety, compute.

11

STAGE 11 AI GOVERNANCE ANSWER SPINE

AI GOVERNANCE ANSWER SPINE · METHOD, MODEL, USE CASE AND DEPLOYMENT · MISSION PILLAR, MINISTRY, IMPLEMENTER AND REGULATOR · DATA, MODEL, PROCUREMENT, DECISION AND MONITORING · HUMAN REASONS, CONTESTABILITY AND REMEDY · OUTLAY, COMPUTE, MODEL AND SUMMIT STATUS OFFICIALLY · EVIDENCE LIMIT · STATE MODEL TYPE, TASK, DATASET/METRIC

DEFINE → method, model, use case and deployment → **MAP → mission pillar, ministry, implementer and regulator** → **TRACE → data, model, procurement, decision and monitoring** → **RETAIN → human reasons, contestability and remedy** → **VERIFY → outlay, compute, model and summit status officially**

CONCEPT / ARCHITECTURE
- DEFINE → method, model, use case and deployment
- MAP → mission pillar, ministry, implementer and regulator

MECHANISM / APPLICATION
- TRACE → data, model, procurement, decision and monitoring
- RETAIN → human reasons, contestability and remedy

READINESS / STATUS LIMIT
- VERIFY → outlay, compute, model and summit status officially
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State model type, task, dataset/metric,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State model type, task, dataset/metric,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State model type, task, dataset/metric,.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · INDIA'S AI BOTTLENECK IS NOT ONLY RESEARCH QUALITY · IT IS ALSO COMPUTE ACCESS, MULTILINGUAL DATA, EVALUATION FRAMEWORKS · A PRO-INNOVATION STANCE DOES NOT MEAN REGULATORY VACUUM · IT USUALLY MEANS GUIDELINES, STANDARDS, INSTITUTIONAL OVERSIGHT AND SECTOR-SENSITIVE · INDIA'S GOVERNANCE CHALLENGE IS SHAPED BY SCALE, MULTILINGUALITY, WELFARE-STATE · CENTRAL MINISTRY ANCHOR. · INDIAAI MISSION

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH
- CAUTION: India's AI bottleneck is not only research quality
- it is also compute access, multilingual data, evaluation frameworks and diffusion into public-interest use cases.
- CAUTION: A pro-innovation stance does not mean regulatory vacuum
- it usually means guidelines, standards, institutional oversight and sector-sensitive calibration.

READINESS, UNIT OR STATUS LIMIT
- CAUTION: India's governance challenge is shaped by scale, multilinguality, welfare-state interaction and uneven digital capacity.
- MeitY: central ministry anchor.
- IndiaAI Mission: Cabinet-approved mission structure.
- Digital India AI initiative: broad public-facing programme explanation.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE
- India AI Governance Guidelines page: official mission-linked governance framing.
- Safe & Trusted AI pillar: official IndiaAI governance/safety pillar.
- AI can strengthen public services, agriculture, health, education, citizen interfaces and productivity tools.
- Indigenous compute and model-development capacity matter for strategic autonomy in language technologies and sensitive sectors.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels